Heart Disease Prediction Using Machine Learning

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Project Title: Heart Disease Prediction

Abstract

This project aims to predict the presence of heart disease using Machine Learning techniques. Logistic Regression was used to train a predictive model on a heart disease dataset. The model was evaluated using Accuracy, Precision, Recall, and F1 Score metrics.

Introduction

Heart disease is one of the leading causes of death worldwide. Early detection can help healthcare professionals provide timely treatment. Machine Learning techniques can assist in predicting heart disease based on patient medical information.

Objectives

- Load and analyze heart disease data
- Preprocess the dataset
- Train a Machine Learning model
- Predict heart disease
- Evaluate model performance

Dataset Description

The dataset contains medical information such as:

- Age
- Sex
- Chest Pain Type
- Resting Blood Pressure

- Cholesterol
- Maximum Heart Rate
- Other medical attributes

Target Variable:

- 0 = No Heart Disease
- 1 = Heart Disease

Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Visual Studio Code
- GitHub

Methodology

Step 1: Load Dataset

The dataset was loaded using Pandas.

Step 2: Feature Selection

Input features and target variable were separated.

Step 3: Data Splitting

The dataset was divided into training and testing sets.

Step 4: Model Training

A Logistic Regression model was trained using the training dataset.

Step 5: Prediction

Predictions were generated on the testing dataset.

Step 6: Evaluation

The model was evaluated using:

- Accuracy

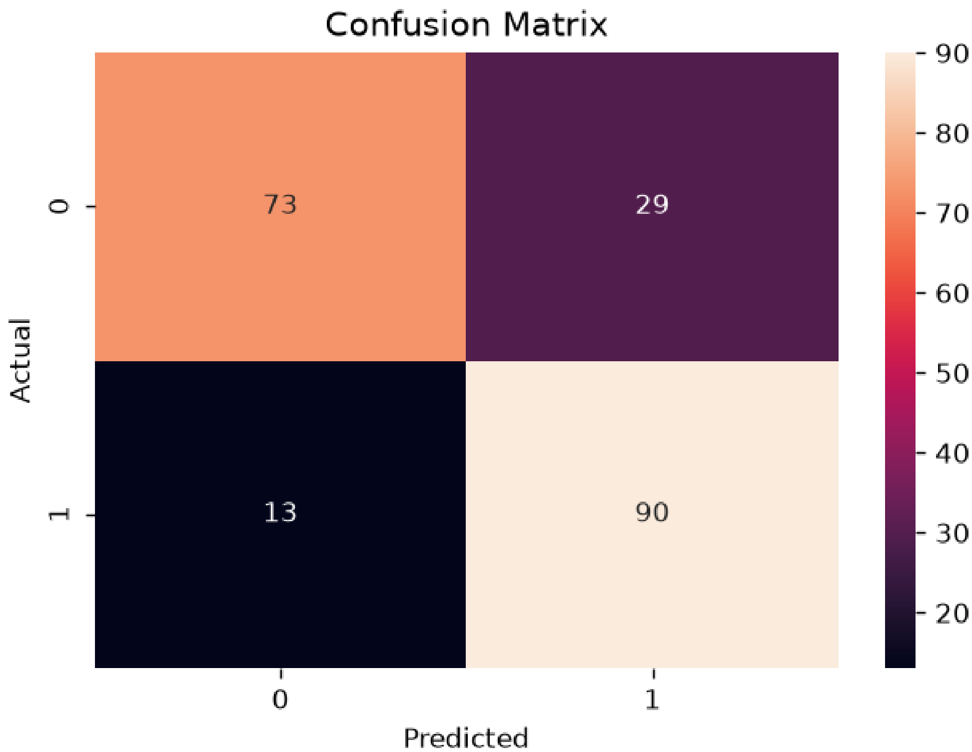
- Precision
- Recall
- F1 Score
- Confusion Matrix

Results

Metric	Value
Accuracy	79.51%
Precision	75.63%
Recall	87.38%
F1 Score	81.08%

Confusion Matrix:

```
[[73 29]
 [13 90]]
```



Conclusion

The Logistic Regression model successfully predicted heart disease with an accuracy of approximately 80%. The results indicate that Machine Learning can be effectively applied to healthcare datasets for disease prediction.

References

1. Scikit-Learn Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Heart Disease Dataset